

Quantum_Computing_Lab_Assignment_2

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[28]: !pip -q install qiskit
!pip -q install qiskit-ibm-runtime
!pip -q install qiskit_aer
!pip -q install qiskit[visualization]
!pip -q install pylatexenc
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
from qiskit import QuantumCircuit, transpile
from math import pi, sqrt
from qiskit.visualization import plot_bloch_multivector, plot_histogram
import matplotlib.pyplot as plt
```

1 Q1: Explain the teleportation concept in detail by the experiment.

1.1 Explanation:

Quantum teleportation transfers a qubit's state $|\psi\rangle$ from Alice to Bob without sending the physical qubit.

We use: - **q0**: Alice's qubit with state $|\psi\rangle$ (prepared here using an X gate for $|1\rangle$ state).

- **q1**: Alice's half of an entangled pair.
- **q2**: Bob's half of an entangled pair.

Steps: 1. Prepare $|\psi\rangle$ on q0.

2. Create an entangled pair between q1 and q2 (H on q1, CX(q1→q2)).
3. Alice does a Bell measurement on q0 and q1 (CX(q0→q1), H(q0), measure q0, q1).
4. Based on Alice's two classical bits, Bob applies corrections to q2:
 - If second bit = 1 → Apply X to q2 (CX(q1→q2) simulates this in our circuit).
 - If first bit = 1 → Apply Z to q2 (CZ(q0→q2) simulates this).
5. Measure q2 — this should match the original $|\psi\rangle$.

```
[25]: %matplotlib inline
import matplotlib.pyplot as plt

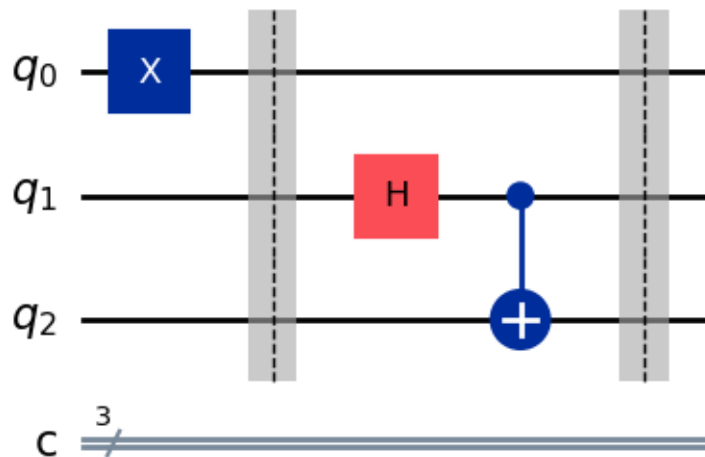
# Create circuit with 3 qubits and 3 classical bits
qc = QuantumCircuit(3, 3)

# Prepare  $|\psi\rangle$  on  $q_0$  (here  $|1\rangle$ )
qc.x(0)
qc.barrier()

# Create entanglement between  $q_1$  and  $q_2$ 
qc.h(1)
qc.cx(1, 2)
qc.barrier()

qc.draw('mpl')
```

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2 Q2. Creating a circuit that works on a Quantum Register with 3 Qubits.

2.1 Explanation:

We now add the Bell measurement part and show how Alice's two classical bits are obtained.

Process:

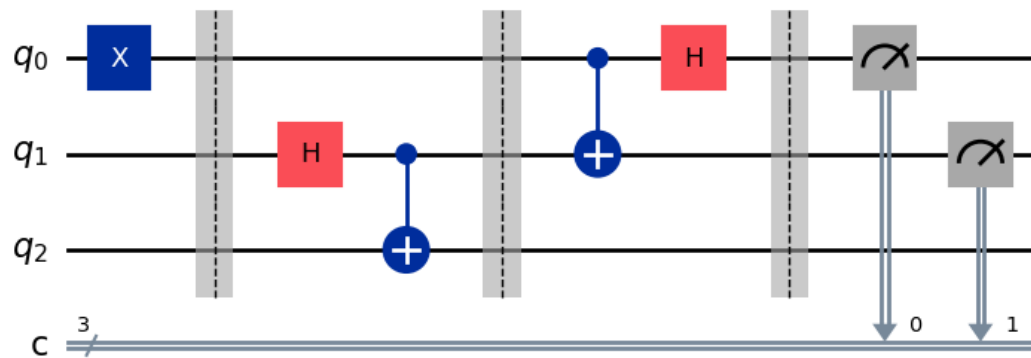
- Apply $CX(q_0 \rightarrow q_1)$ and $H(q_0)$ for Bell measurement.
- Measure q_0 and q_1 into classical bits c_0 and c_1 .

```
[26]: # Bell measurement
qc.cx(0, 1)
qc.h(0)
qc.barrier()

# Measure Alice's qubits into c0 and c1
qc.measure([0, 1], [0, 1])

qc.draw('mpl')
```

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Q3. Write a code for the measurement of the Circuit

2.2 Explanation:

Finally, we apply Bob's corrections based on Alice's bits and measure Bob's qubit:

- CX($q_1 \rightarrow q_2$) applies X to q_2 if $c_1 = 1$.
- CZ($q_0 \rightarrow q_2$) applies Z to q_2 if $c_0 = 1$.
- Measure q_2 to verify teleportation.

Then run the circuit on a simulator and plot the measurement histogram.

```
[27]: # Bob's conditional corrections
qc.cx(1, 2)
qc.cz(0, 2)
qc.barrier()

# Measure Bob's qubit
qc.measure([2], [2])
qc.draw('mpl')

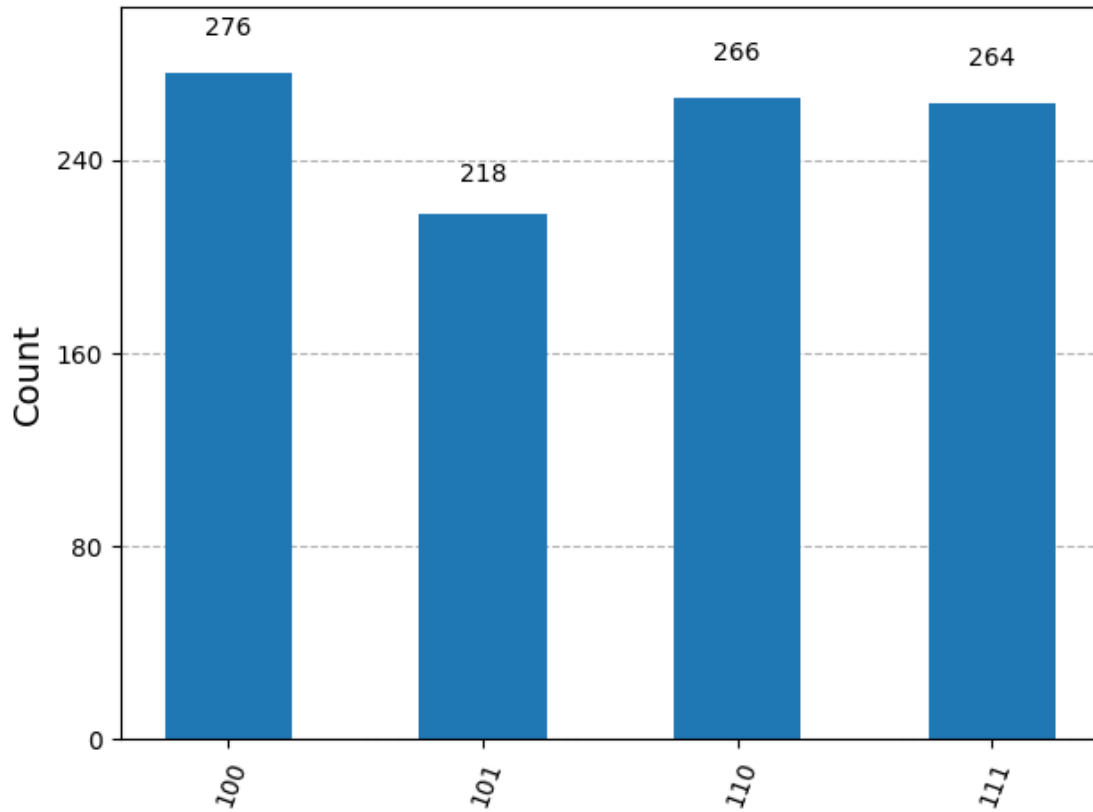
# Run the circuit
```

```
backend = AerSimulator(method="automatic")
circuit = transpile(qc, backend)
result = backend.run(circuit, shots=1024).result()
count = result.get_counts()

print("Measurement results:", count)
plot_histogram(count)
```

Measurement results: {'110': 266, '100': 276, '101': 218, '111': 264}

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